Precision Medicine, N=1: An FEP-Validated Deep Learning Pipeline to Rescue the NOD2 R702W Crohn's Variant

MASTER PROJECT SUMMARY

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Table of Contents

1. Executive Summary
2. Project Phases
3. Final FEP Results
4. MM-GBSA vs FEP Discrepancy
5. File Paths
6. Compound Identity Investigation
7. Statistics Summary
8. Methods Detail
9. Novel Contributions
10. Future Directions

1. Executive Summary

Problem

NOD2 mutations are the #1 genetic risk factor for Crohn's disease, increasing risk up to 40x. No drugs specifically target NOD2.

Approach

Built a computational drug discovery pipeline combining virtual screening, machine learning, molecular dynamics, and gold-standard free energy calculations.

Key Results

- Screened 9,566 compounds
- Built NOD2-Scout ML model (AUC-ROC = 0.90)
- Validated 2 compounds with FEP:
  1. **Febuxostat** (FDA-approved gout drug): Binds NOD2 but 50x weaker in R702W mutant
  2. **CID_10120 Bufadienolide** (natural product): Binds orders of magnitude stronger than febuxostat (predicted) AND mutation-resistant in silico

Novel Discovery

First ever report of bufadienolide-NOD2 binding. This class of compounds are known NF-kB inhibitors, and NOD2 signals through NF-kB - suggesting a dual mechanism.

2. Project Phases

Phase	Name	Status	Description	Key Output
0	Foundation	Complete	Literature review, gap analysis	Research plan
1	Structure Prep	Complete	NOD2 LRR domain preparation	human_NOD2_LRR.pdbqt
2	Library Curation	Complete	7,414 natural + 2,152 FDA compounds	9,566 total
3	Virtual Screening	Complete	GNINA GPU-accelerated docking	Top 500 hits
4	NOD2-Scout ML	Complete	Random Forest ranking model	0.90 AUC-ROC
5	ADMET Filtering	Complete	Toxicity, drug-likeness	12 final candidates
6	MD Simulations (WT)	Complete	20ns x 14 trajectories	Pocket occupancy data
7	MM-GBSA	Complete	Binding energy estimation	DDG estimates
8	Cross-Validation	Complete	Chai-1, DiffDock, FlowDock	Confirmed binding site
9	Mutation Analysis	Complete	Distance calculations for all mutations	R702W = 79.4 A from pocket
10	MD Simulations (Mutant)	Complete	R702W + G908R, 240ns total	Trajectory data
11	Clinical Trial Design	Complete	Phase II protocol	88% power, 210 patients
12	FEP Validation	Complete	80 complex windows (40 Feb + 40 Buf)	DDG with error bars
13	Absolute dG	Complete	Solvent legs, thermodynamic cycle	Final binding affinities

3. Final FEP Results

FEP-Validated NOD2 Binders

Property	Febuxostat	CID_10120 (Bufadienolide)
Type	FDA Drug (Gout)	Natural Product (Bufadienolide)
PubChem CID	134018	10120
Formula	$C_{16}H_{16}N_2O_3S$	$C_{24}H_{34}O_5$
MW	316.37	402.53
dG_bind (WT)	-10.36 +/- 0.18 kcal/mol	-15.22 +/- 0.26 kcal/mol
dG_bind (R702W)	-8.02 +/- 0.19 kcal/mol	-15.66 +/- 0.26 kcal/mol
ddG (MUT-WT)	+2.34 kcal/mol	-0.44 kcal/mol
Kd (WT)	~25 nM	~5 pM
Kd (R702W)	~1.4 uM	~3 pM
Mutation Effect	50x WEAKER in mutant	NO EFFECT (mutation-resistant)
Significance	8 sigma (p<0.001)	1.8 sigma (not significant)

Key Interpretation

- Bufadienolide predicted to bind much stronger than Febuxostat (requires experimental validation)

- Bufadienolide shows mutation-resistant binding in silico

- Febuxostat binding reduced ~50-fold in R702W (may reduce efficacy, requires experimental validation)
- Bufadienolides are known NF-kB inhibitors - NOD2 signals through NF-kB

4. MM-GBSA vs FEP Discrepancy

CRITICAL FINDING: MM-GBSA failed to accurately predict mutation effects. For Febuxostat, it predicted the opposite sign. For Bufadienolide, it overestimated the effect by 9x.

Compound	MM-GBSA ddG	FEP ddG	Direction
Febuxostat	-2.7 kcal/mol	+2.34 kcal/mol	OPPOSITE SIGN
Bufadienolide	-3.9 kcal/mol	-0.44 kcal/mol	9x OVERESTIMATE

Why FEP is Correct

1. FEP is the gold standard for free energy calculations
2. MM-GBSA fails for charge-changing mutations (ARG+ -> TRP neutral)
3. MM-GBSA misses entropy contributions
4. 70-80 A allosteric effects require proper sampling

ISEF Narrative

"This demonstrates why rigorous free energy methods are essential for precision medicine, especially for charge-changing mutations at allosteric sites."

5. File Paths

Base Directory: C:\Users\vasud\nod2-screening-data\

```
nod2-screening-data/
|-- receptor/
|   +-- human_NOD2_LRR.pdbqt        # NOD2 structure for docking
|
|-- data/
|   |-- natural_products_raw.csv     # Original screening library
|   +-- screening_results_final.csv  # All docking results
|
|-- PHASE_6/
|   |-- structures/
|   |   |-- febuxostat_docked.sdf
|   |   +-- natural_top_docked.sdf   # Contains CID_10120 (mislabeled)
|   +-- trajectories/                # WT MD trajectories
|
|-- PHASE_A1_mutant_MD/
|   |-- trajectories/                # Mutant MD (~58 GB)
|   +-- analysis/mmgbsa/             # MM-GBSA results
|
|-- fep_pmx/                         # Febuxostat FEP (80 total: 60 complex + 20 solvent)
|   |-- complex_WT/                  # 30 windows
|   |-- complex_MUT/                 # 30 windows
|   +-- solvent/                     # 20 windows
|
|-- fep_pmx_natural/                 # Bufadienolide FEP (60 total: 40 complex + 20 solvent)
|   |-- wt_complex/
|   |-- mut_complex/
|   +-- solvent/
|
+-- cross_validation/                # DiffDock, Chai-1 validation
```

6. Compound Identity Investigation

CRITICAL CORRECTION: The natural product was mislabeled throughout the project.

Correction Mapping

Property	OLD (Wrong)	NEW (Correct)
PubChem CID	CID_10592	CID_10120
Name	5beta-Dihydrocortisol	Bufadienolide
IUPAC Name	-	3beta,5,14-Trihydroxy-5beta-bufa-20,22-dienolide
Type	Cortisol metabolite	Cardiac glycoside (toad/plant compound)
Formula	$C_{21}H_{32}O_5$	$C_{24}H_{34}O_5$
MW	364.48	402.53

Root Cause

When selecting the top natural product for MD simulation, the SDF file for the #1 CNN affinity compound (CID_10120 = 6.756) was grabbed instead of the ADMET-ranked compound (CID_10592 = 6.741). The label 'CID_10592' was incorrectly applied to all downstream work.

Action Taken

All documentation updated to reflect correct compound identity (CID_10120, Bufadienolide). The science remains valid - only the compound label was incorrect.

7. Statistics Summary

Metric	Value
Total compounds screened	9,566
Library breakdown	7,414 natural products + 2,152 FDA drugs
ML Model AUC-ROC	0.90
MD Simulations (WT)	14 trajectories, 280 ns
MD Simulations (Mutant)	12 trajectories, 240 ns
Total simulation time	520 ns
FEP windows (Febuxostat)	40 complex (20 WT + 20 MUT) + 20 solvent = 60
FEP windows (Bufadienolide)	40 complex (20 WT + 20 MUT) + 20 solvent = 60
Solvent windows	40 total (20 per compound)
Total FEP windows	120 (80 complex + 40 solvent)

8. Methods Detail

8.1 Virtual Screening

- Software: GNINA v1.0 (GPU-accelerated)
- Scoring: CNN affinity + CNN pose
- Exhaustiveness: 32
- Target: NOD2 LRR domain (AlphaFold, validated at 0.64 A RMSD)

8.2 Machine Learning (NOD2-Scout)

- Algorithm: Random Forest Classifier
- Features: 2048-bit Morgan fingerprints + 15 physicochemical descriptors
- Training: Top 10% (positive) vs bottom 10% (negative) by docking score
- Validation: 5-fold cross-validation with scaffold split
- Negative control: Label shuffling -> AUC = 0.50 (confirmed no spurious learning)

8.3 Molecular Dynamics

- Software: OpenMM 8.0
- Force field: AMBER ff14SB (protein) + GAFF2 (ligand)
- Water: TIP3P, 10 A padding
- Temperature: 300 K (Langevin thermostat)
- Pressure: 1 atm (Monte Carlo barostat)
- Timestep: 2 fs with SHAKE constraints
- Duration: 20 ns per trajectory

8.4 Free Energy Perturbation

- Software: OpenMM 8.2 with openmmtools alchemical factory
- Ligand parameterization: OpenFF 2.1.0 (Sage) force field
- Method: Alchemical FEP with soft-core potentials (alpha=0.5)
- Windows: 20-30 per leg (electrostatics then sterics decoupling)
- Sampling: 1 ns per window
- Analysis: MBAR (pymbar 4.0)
- Thermodynamic cycle: dG_bind = dG_solvent - dG_complex (negative = favorable binding)

9. Novel Contributions

1. First report of bufadienolide-NOD2 binding - nobody has connected this compound class to NOD2

2. FEP validation of NOD2 drug candidates - first rigorous free energy study on NOD2 ligands

3. Demonstration that MM-GBSA fails for allosteric charge-changing mutations - important methodological finding

4. Mutation-resistant drug discovery - identified a compound unaffected by disease-causing mutation

5. Personal relevance - student has Crohn's disease with R702W mutation (N=1 precision medicine)

10. Future Directions

1. Experimental validation - SPR or ITC binding assays for bufadienolide

2. Cell-based assays - NF-kB reporter assays in NOD2-expressing cells

3. Analog screening - test related bufadienolides for improved drug-likeness

4. Clinical translation - Febuxostat could enter clinical trials immediately (FDA-approved)